Predicting Customer Churn

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Agenda

Intro & Exploratory Data Analysis

- Tyler

Model Selection & Performance

- Sahas & Nihal

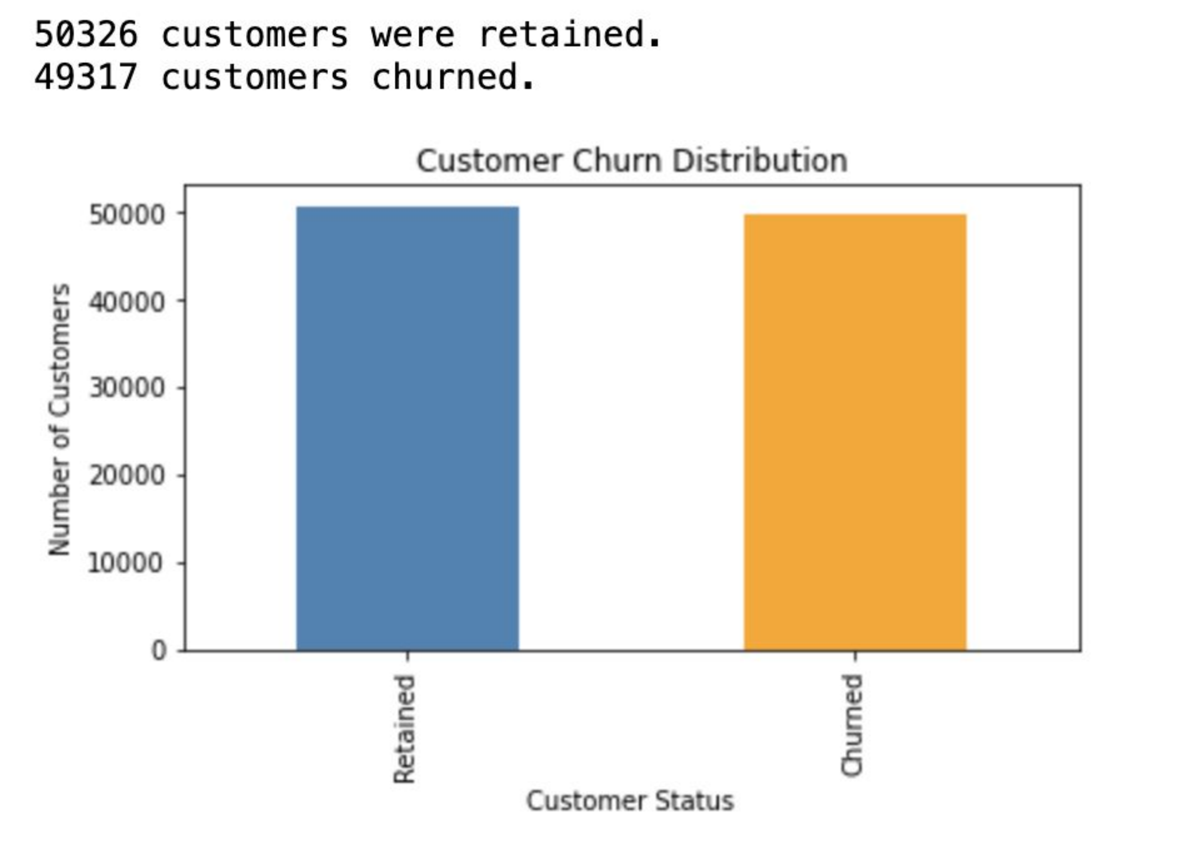
Model Optimization

- James

Conclusion

- Laine

The Dataset



- 100,000 records - each indicative of a single telecom customer

- 100 attributes per customer
  - 81 Numerical
  - 17 Categorical

- Attributes relate to a customer's:
  - Usage Metrics
  - Revenue
  - Call Behavior
  - Demographics
  - Equipment Details

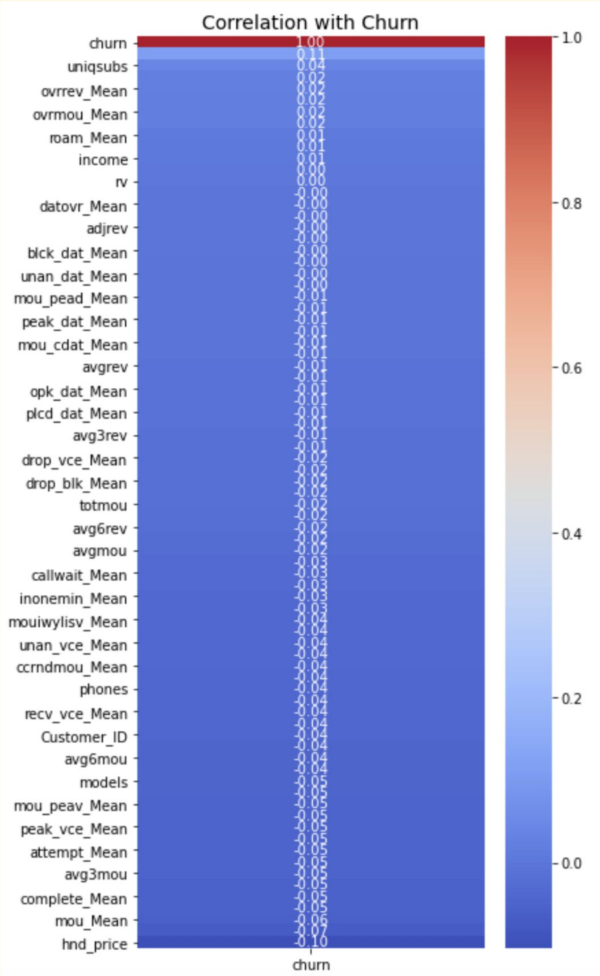
Churn Ratio for Categorical Features

	feature	category	churn_rate
1	crclscod	A2	0.615563
10	ethnic	O	0.580753
4	area	NORTHWEST/ROCKY MOUNTAIN AREA	0.569085
7	hnd_webcap	WC	0.563028
5	dualband	N	0.536817
6	refurb_new	R	0.534361
3	prizm_social_one	R	0.527068
11	kid0_2	Y	0.519442
8	marital	U	0.512656
2	asl_flag	N	0.509598
16	creditcd	N	0.508281

Churn Ratio for Numerical Features

	feature	bin_range	churn_rate
66	hnd_price	(9.989, 29.99]	0.574615
73	eqpdays	(530.0, 1823.0]	0.557248
48	months	(11.0, 16.0]	0.554434
2	totmrc_Mean	(-26.916, 30.0]	0.549672
41	mou_opkv_Mean	(-0.001, 18.537]	0.548396
28	mou_cvce_Mean	(-0.001, 49.051]	0.547360
37	mou_peav_Mean	(-0.001, 37.642]	0.544560
1	mou_Mean	(-0.001, 150.75]	0.543919
45	complete_Mean	(-0.001, 28.667]	0.542092
21	comp_vce_Mean	(-0.001, 28.667]	0.541968
39	opk_vce_Mean	(-0.001, 10.333]	0.541649

Exploratory Data Analysis



- No individual feature has a strong correlation with churn

- Top Correlated features
    - eqpdays (.11)
    - hnd_price (-0.10)

Model Selection Strategy

- Shortlisted for data set: Logistic Regression, Random Forest, Gradient Boosting
  - Ability to perform well on categorical/numerical data
  - Efficient for classification problems

- Evaluation Criteria: ROC AUC (Test), AUC Gap, Training Time, Accuracy

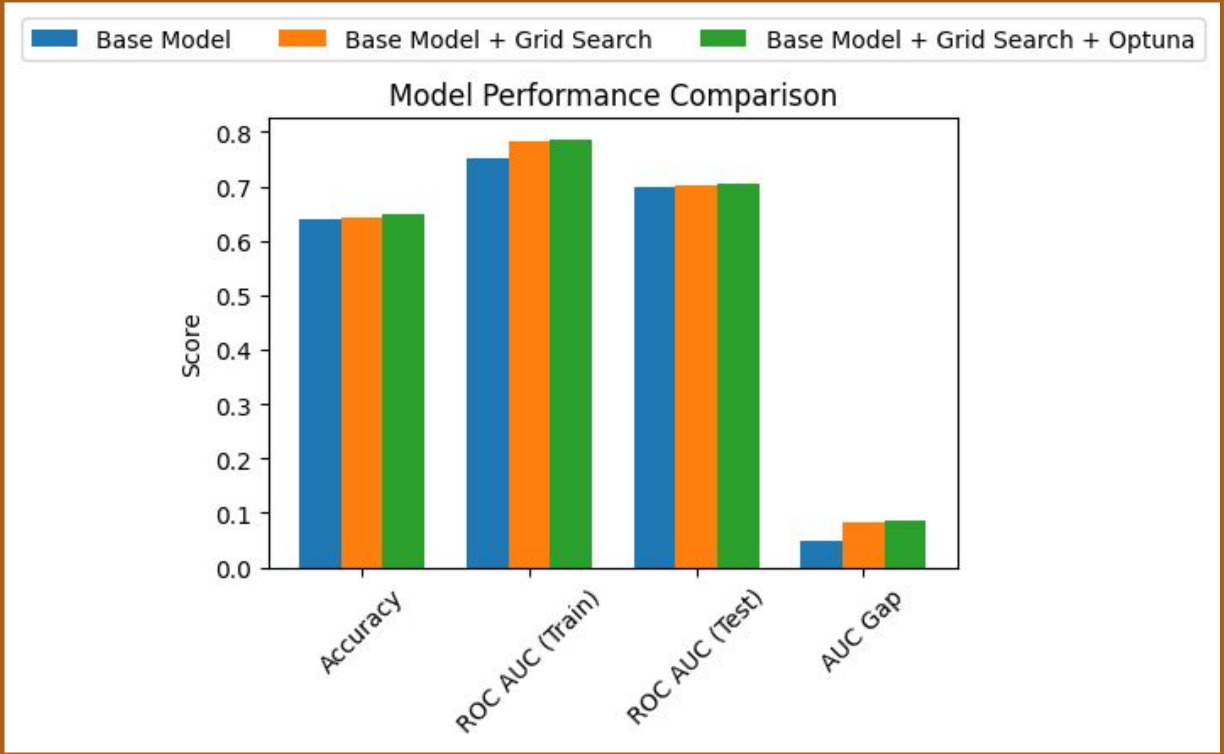
Benchmarking Process:

1. Feature Engineering
2. Preprocessing Pipelines
3. Get a baseline result for each of the models
4. Try various gradient boosting models
5. Evaluate

=== Benchmarking Results ===

Model	Accuracy	ROC AUC (Train)	ROC AUC (Test)	AUC Gap	Training Time (s)
LightGBM	0.6440	0.7480	0.6967	0.0513	12.84
XGBoost	0.6302	0.8394	0.6858	0.1535	22.31
GradientBoosting	0.6311	0.6913	0.6832	0.0081	382.17

Model Performance Comparison



- Slight Improvement with Tuning
- Slight Increase in AUC Gap From around 0.75 to 0.79
- Accuracy Remains Stable at around 64% across all models

Partial Dependence Plots

1. eqpdays

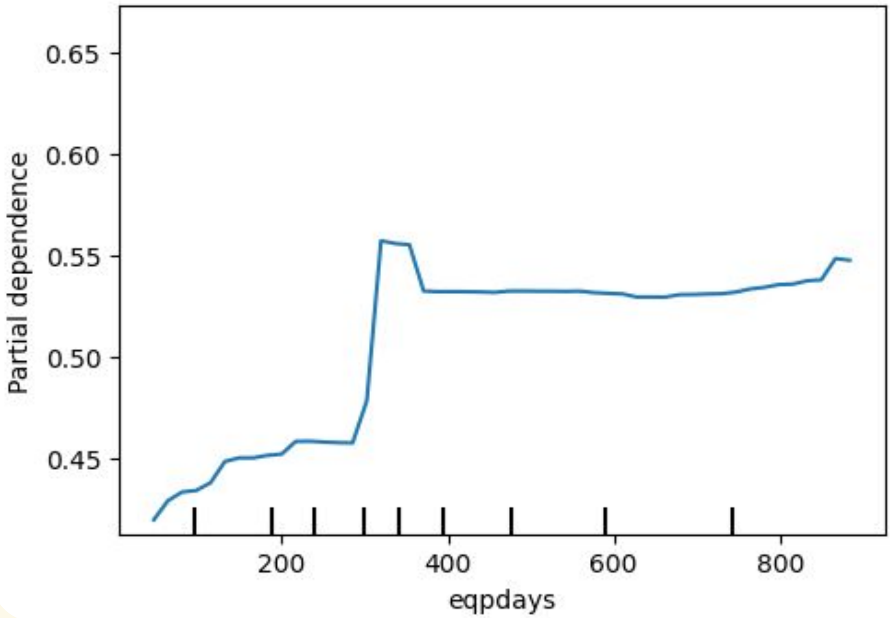
Between 300 days and 400 days we observe a positive swing in churn probability, likely due to annual contracts

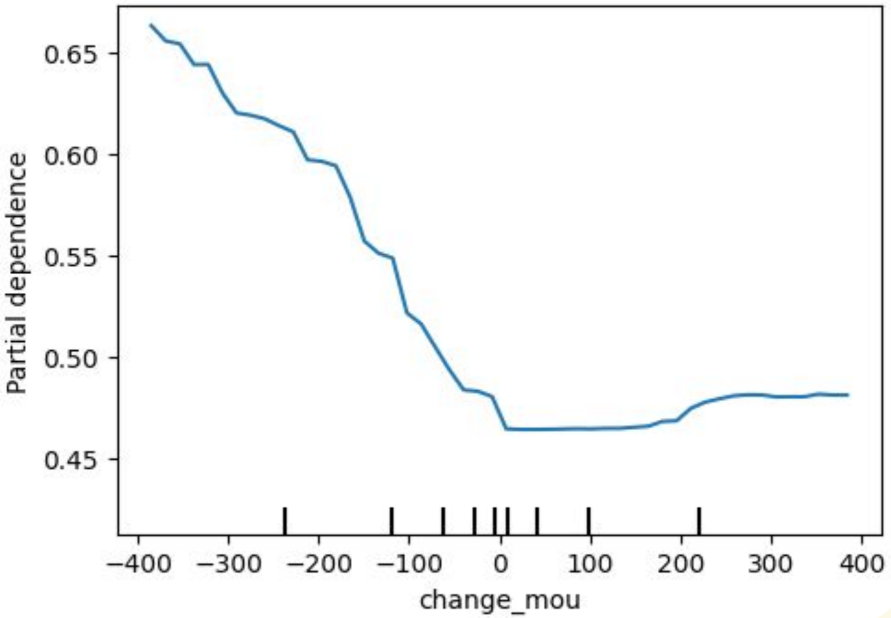
2. change_mou

Large debt payments are bad for business. Keep the customer on a monthly paid payment cycle

3. General Observation

Or model maybe in the 70% accuracy range however we can find patterns in the data to help control the churn for the future





Key Findings/Predictive Insights

Strongest Indicators of Churn

- Change in Usage (change_mou): One of the most important predictors across SHAP analysis and feature importance rankings. A drop in usage is a strong indicator of churn.

- Average Usage (mou_Mean): Lower engagement (reflected in average usage) is associated with higher churn risk.

- Monthly charges (totmrc_Mean): Pricing tiers and monthly spending levels influence churn. Lower or highly variable charges are linked with higher churn.

- Customer Tenure (months): Shorter-tenure customers are more likely to churn, as confirmed by SHAP and correlation analysis.

- Device Age(eqpdays): Customers with older devices showed higher churn likelihood, suggesting lifecycle or dissatisfaction effects.

Strategic Recommendations & Next Steps

Proactive Customer Retention

- **Monitor usage drop-offs**: Set alerts for significant decreases in change_mou and change_rev to trigger outreach.
- **Targeted offers**: Deploy discounts or loyalty rewards for customers with declining mou_Mean or nearing key lifecycle stages (eqpdays, months).
- **Short-tenure focus**: Develop onboarding retention programs for customers within their first 6–12 months.

Lifecycle Optimization

- **Upgrade incentives**: Encourage device upgrades around 300–400 days (eqpdays) to reduce churn tied to outdated tech.
- **Contract alignment**: Time promotions around annual contract anniversaries to preempt churn risk.
- **Customer segmentation**: Use tenure (months) and engagement features to create risk-based lifecycle campaigns.

Future Enhancements

- **Real-time integration**: Implement streaming data pipelines for change_mou, totmrc_Mean, and usage metrics to improve responsiveness.
- **Model updates**: Schedule quarterly performance reviews with updated training data to combat data drift.
- **SHAP-based explainability**: Use SHAP outputs to personalize retention messaging based on key churn drivers per customer.

Questions?

THANK YOU

Model Selection

```
LightGBM Accuracy: 0.64
[[6282 3739]
 [3403 6576]]
              precision    recall  f1-score   support

           0       0.65      0.63      0.64     10021
           1       0.64      0.66      0.65      9979

    accuracy                           0.64     20000
   macro avg       0.64      0.64      0.64     20000
weighted avg       0.64      0.64      0.64     20000
```

```
Gradient Boosting + Feature Engineering + Feature Selection
Accuracy: 0.63
ROC AUC: 0.68
```

```
LightGBM + Feature Selection Accuracy: 0.64
[[6268 3753]
 [3437 6542]]
              precision    recall  f1-score   support

           0       0.65      0.63      0.64     10021
           1       0.64      0.66      0.65      9979

    accuracy                           0.64     20000
   macro avg       0.64      0.64      0.64     20000
weighted avg       0.64      0.64      0.64     20000
```

```
XGBoost + Feature Engineering + Feature Selection
Training Time: 24.70 seconds
Accuracy: 0.63
ROC AUC (Train): 0.84
ROC AUC (Test):  0.69
AUC Gap (Overfitting Signal): 0.15
```

```
LightGBM + Feature Selection + Feature Engineering
Accuracy: 0.64
AUC Score: 0.70
```

	feature	importance
9	change_mou	198
1	mou_Mean	183
2	totmrc_Mean	149
42	months	142
10	change_rev	134
64	eqpdays	121
53	avgqty	88
52	avgmou	88
60	hnd_price	69
23	mou_cvce_Mean	68
11	drop_vce_Mean	68
51	avgrev	60
38	drop_blk_Mean	55
43	uniqsubs	49
36	mou_opkv_Mean	45
32	mou_peav_Mean	44
0	rev_Mean	44
56	avg3rev	39
30	peak_vce_Mean	39
4	ovrmou_Mean	39

- change_mou
  - Customers with large drops in usage are churning more — big signal!
- mou_Mean
  - Overall minutes of use is a strong predictor — likely correlated with engagement
- totmrc_Mean
  - Average recurring monthly charge—pricing tier/plan type effect
- months
  - Tenure: short-tenure customers tend to churn more
- change_rev
  - Recent drop in spend—similar idea to change_mou
- eqpdays
  - Days since getting phone—device age matters (likely loyalty/lifecycle)